String Problems Sequence (Easy → Hard)

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1. Remove Duplicate Characters from a String

https://www.geeksforgeeks.org/dsa/remove-duplicates-from-a-given-string/

Problem:
Given a string, remove all duplicate characters while keeping only the first occurrence of each character.

Example:

text
```
Input:  "programming"
Output: "progamin"

Input: "banana"
Output: "ban"
```

```
public class Main {
    public static void main(String[] args) {
        String input = "programming"; // Change to "banana" for the other test
        String result = "";

        for (int i = 0; i < input.length(); i++) {
            char currentChar = input.charAt(i);

            // Check if currentChar is already in result
            boolean found = false;
            for (int j = 0; j < result.length(); j++) {
                if (result.charAt(j) == currentChar) {
                    found = true;
                    break;
                }
            }

            // If currentChar not found in result, add it
            if (!found) {
                result = result + currentChar;
            }
        }

        System.out.println("Input: " + input);
```

```
        System.out.println("Output: " + result);
    }
}
```

Example outputs:

- For `"programming"` ⇒ `progamin`
- For `"banana"` ⇒ `ban`

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2. Substring Check

https://www.geeksforgeeks.org/dsa/java-program-to-check-if-a-string-is-substring-of-another/

Problem:
Given two strings `mainStr` and `subStr`, check if `subStr` exists inside `mainStr`.

Example:

text
```
Input:
mainStr = "hello world"
subStr  = "world"
Output: true

Input:
mainStr = "java"
subStr  = "python"
Output: false
```

```java
public class Main {
    public static void main(String[] args) {
        String mainStr = "hello world";
        String subStr = "world";
        boolean found = false;

        // loop over each possible starting index in mainStr
  for (int i = 0; i <= mainStr.length() - subStr.length(); i++) {
            boolean match = true;
            // check if substring starting at i matches subStr
            for (int j = 0; j < subStr.length(); j++) {
                if (mainStr.charAt(i + j) != subStr.charAt(j)) {
                    match = false;
                    break;
                }
            }
            if (match) {
                found = true;
                break;
          }
        }

        System.out.println("mainStr = " + mainStr);
        System.out.println("subStr = " + subStr);
        System.out.println("Output: " + found);
    }
}
```

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Alternative solution of problem 2

```java
public class Main {
    public static void main(String[] args) {
        String mainStr = "hello world"; // Try "java" and "python" for another test
        String subStr = "world";
        boolean found = false;

        int i = 0;
        while (i <= mainStr.length() - subStr.length()) {
            int j = 0;
```

```
            // Compare character by character while they match
            while (j < subStr.length() && mainStr.charAt(i + j) == subStr.charAt(j)) {
                j++;
            }

            if (j == subStr.length()) {
                found = true;
                break;
            }
            i++; // Try next position
        }

        System.out.println("mainStr = " + mainStr);
        System.out.println("subStr = " + subStr);
        System.out.println("Output: " + found);
    }
}
```

How it works

- Slide over each position of `mainStr` where `subStr` could fit.
- For each possible starting index `i`, check character by character if all chars in `subStr` match those at position `(i + j)` in `mainStr`.
- If match found, set `found` to `true`.

Example outputs

1.

mainStr = "hello world"
subStr = "world"
Output: true

2.

mainStr = "java"
subStr = "python"
Output: false

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3. Find the Smallest Word in a String

https://www.geeksforgeeks.org/dsa/program-find-smallest-largest-word-string/

Problem:
Given a sentence, find the smallest (shortest) word. If there are multiple, return the first one.

Example:

text
```
Input: "I love coding"
Output: "I"

Input: "cat is cute"
Output: "is"
```

```
public class Main {
    public static void main(String[] args) {
        String input = "cat is cute";  // Try "I love coding" for other example

        String smallest = "";   // Holds the smallest word found so far
        String tempWord = "";   // Build current word character by character

        for (int i = 0; i <= input.length(); i++) {
            // If we're inside string, and not on a space, add to tempWord
            if (i < input.length() && input.charAt(i) != ' ') {
                tempWord = tempWord + input.charAt(i);
            } else {
                // We just finished a word (i is space or end)
                if (tempWord.length() > 0) {
               // If first word, or this word is smaller than current smallest
               if (smallest.equals("") || tempWord.length() < smallest.length()) {
                            smallest = tempWord;
                    }
                    // Reset tempWord to start next word
                    tempWord = "";
                }
            }
        }

        System.out.println("Input: " + input);
        System.out.println("Output: " + smallest);
    }
}
```

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4. Find the Longest Word in a String

https://www.geeksforgeeks.org/dsa/program-display-number-characters-longest-word-sentence/

Problem:
Given a sentence, find the longest word. If there are multiple, return the first one.

Example:

text
```
Input: "I love programming"
Output: "programming"

Input: "the sun shines bright"
Output: "shines"
```

```
public class Main {
    public static void main(String[] args) {
        String input = "the sun shines bright";  // Try "I love programming" for other example

        String longest = "";      // To hold the longest word found
        String tempWord = "";     // To build the current word

        for (int i = 0; i <= input.length(); i++) {
            // If not space or end of string, build the word
            if (i < input.length() && input.charAt(i) != ' ') {
                tempWord = tempWord + input.charAt(i);
            } else {
                // At space or end of string, end of a word
                if (tempWord.length() > 0) {
                    if (longest.equals("") || tempWord.length() > longest.length()) {
                        longest = tempWord;
                    }
                    tempWord = ""; // Reset for next word
                }
            }
        }

        System.out.println("Input: " + input);
        System.out.println("Output: " + longest);
```

```
    }
}
```

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5. Reverse the Words in a String

https://www.geeksforgeeks.org/java/reverse-words-given-string-java/

Problem:
Reverse the order of words without reversing individual characters.

Example:

```
Input: "I love coding"
Output: "coding love I" ; // gnidoc evol I

Input: "one two three"
Output: "three two one"
```

```
public class Main {
    public static void main(String[] args) {
        String input = "one two three"; // Try "I love coding" for other example
        String result = "";  // Will contain the final sentence in reverse word order

        String word = ""; // Temporary word

        // Go from left to right, collect each word
        for (int i = 0; i <= input.length(); i++) {
            // If not space and not end, add character to the current word
            if (i < input.length() && input.charAt(i) != ' ') {
                word = word + input.charAt(i);
            } else {
                // At space or at end: we have a complete word (unless word is empty)
                if (!word.equals("")) {
                    // If result is empty, start with the word
                    if (result.equals("")) {
                        result = word;
                    } else {
                        // Otherwise, put the word before the previous result
                        result = word + " " + result;
                    }
                    word = ""; // Reset for the next word
                }
            }
        }
```

```
        System.out.println("Input: " + input);
        System.out.println("Output: " + result);
    }
}
```

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6. Longest Substring Without Repeating Characters

https://www.geeksforgeeks.org/dsa/length-of-the-longest-substring-without-repeating-characters/

Problem:
Find the longest substring that contains only unique characters.

Example:

text
```
Input: "abcabcbb"
Output: "abc"

Input: "bbbbb"
Output: "b"

Input: "pwwkew"
Output: "wke"
```

```
public class Main {
    public static void main(String[] args) {
        String input = "pwwkew"; // Try "abcabcbb" or "bbbbb"
        String maxSub = "";      // Holds the longest substring so far

        // Start at every character in the string
        for (int i = 0; i < input.length(); i++) {
            String tempSub = ""; // Build up substring from i

            for (int j = i; j < input.length(); j++) {
                char c = input.charAt(j);

                // Check if c already exists in tempSub
                boolean repeated = false;
                for (int k = 0; k < tempSub.length(); k++) {
                    if (tempSub.charAt(k) == c) {
```

```
                    repeated = true;
                    break;
                }
            }

            if (repeated) {
                break; // Stop this substring, found a repeat
            } else {
                tempSub = tempSub + c; // Add unique char
            }
        }

        // If the substring built is longer than the max found, update maxSub
        if (tempSub.length() > maxSub.length()) {
            maxSub = tempSub;
        }
    }

    System.out.println("Input: " + input);
    System.out.println("Output: " + maxSub);
  }
}
```

Solution video link :

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7. Find Both the Longest and Smallest Word in a String

https://www.geeksforgeeks.org/dsa/program-find-smallest-largest-word-string/

Problem:
Find the smallest and largest words in a sentence. If there are multiple, return the first one found.

Example:

text

```
Input: "I love programming so much"
Output:
Smallest = "I"
Largest  = "programming"

Input: "this is a test example"
Output:
Smallest = "a"
```

```
Largest  = "example"
```

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